

JUAN BRAIAN HERNÁNDEZ MORANI

AI ENGINEER | FULL STACK DEVELOPER | LLM, RAG & AGENTIC SYSTEMS

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PROFILE

AI engineer and full stack developer building local-first AI and software engineering systems. Creator of SupporterAI Core v1.0-Alpha, integrating local LLM inference, hybrid RAG, vector-backed memory, MCP/tool orchestration, durable project state, and evidence-driven validation. Strong in Python, JavaScript/TypeScript, Java, C#, SQL, and C1-level English.

INDEPENDENT PROJECT EXPERIENCE

SupporterAI - AI Engineer / Full Stack Developer | 2025 - Present | Private R&D

- Build a local-first AI workbench with Python/FastAPI, JavaFX, GGUF/llama.cpp runtimes, Qdrant-backed RAG/memory, and MCP/JSON-RPC integrations.
- Engineer hybrid semantic and lexical retrieval with embeddings, reranking, context pruning, long-term project memory, planning/replanning, and orchestration across 80+ registered tools and capabilities.
- Validate autonomous engineering workflows through 7/7 canonical gates, covering real local-model execution, tool expansion, supervised correction, restart recovery, checkpoints, tests, diffs, and durable continuation.

SAltron - Multimodal Model Systems Research | 2026 - Present | Private R&D

- Design and validate a local-first multimodal architecture spanning 3 primary model systems: cognition, perception/world modeling, and generation, coordinated through asynchronous routing, verification, and resource-aware execution.
- Completed 2 TIES model-merging experiments across Nemotron-based H0 and Qwen-based H1, preserving expected logical behavior in both candidates and multimodal operation in H0, with causal/runtime validation across checkpoints and llama.cpp/Vulkan execution paths.

Selected Software Engineering & AI Projects | 2023 - Present

- OverhaulCrafter: built a unified C#/Unity/BepInEx/Harmony plugin for The Planet Crafter spanning RPG, QoL, NPC, and quest systems; the current release build compiles with 0 warnings and 0 errors; multiple workflows have been validated in-game.
- ZombiesDeOP7DtD: engineered C#/Harmony perception and diagnostics systems using source analysis and decompilation, including physical line of sight, audio occlusion, and deterministic runtime evidence.
- Additional public work includes Storefront Change Guard (local llama.cpp code review), YOLOv11 dual-head training, the PyHelper package published on PyPI, and React/Next.js/Node/PostgreSQL applications.

TECHNICAL SKILLS

AI / LLM: RAG, embeddings, Qdrant/vector databases, semantic + lexical retrieval, reranking, local LLMs (llama.cpp/GGUF), agent/tool orchestration, MCP.

ML / Data: Machine Learning, Deep Learning, PyTorch, PySpark, NumPy, Pandas, statistical analysis.

Backend: Python, FastAPI, Node.js, Express, REST APIs, OAuth/JWT, SQL, PostgreSQL, MySQL.

Frontend: JavaScript, TypeScript, React, Next.js, HTML5, CSS3.

Engineering: Java, C#, Git/GitHub, CI/CD, Linux/WSL2, automated testing, reverse engineering/decompilation, Unity, Harmony, BepInEx.

Languages: Spanish (native), English (C1 - EF SET).

TRAINING & CERTIFICATIONS

AYI Academy - AI & Big Data (Aug 2025 - Nov 2025): LLMs, Machine Learning & Deep Learning | Certificate

Egg LXP - Backend Developer (Mar 2023 - Dec 2023) | **SoyHenry - PERN Full Stack Developer** (May 2022 - Oct 2022) | Certificate

EF SET English Certificate - C1 | Credential